

Sequential Monitoring for Structural Breaks in Heterogeneous Panels: Design, Derivations, and Computational Details

Abstract

This note describes a Monte Carlo study of a real-time structural break monitoring procedure for heterogeneous panels. The procedure works in three steps: (1) each series is pre-filtered by an autoregression estimated on a calibration sample; (2) the filtered residuals are aggregated across series via a pooled pairwise regression; (3) the cumulative sum of squares (CUSQ) of the aggregate residual is compared period by period against a boundary derived from the Brownian bridge, following Wang and Hsiao (2013). The note gives the exact data generating processes, a closed-form derivation showing that the $N(N-1)$ -pair aggregation reduces to a simple cross-sectional average, the two-scenario Monte Carlo design (size under no break, power and detection delay under a known break), and two computational improvements — FFT-based fractional differencing and vectorized AR residuals — that make the full 1.9×10^6 -replication design feasible without changing any statistic.

1 Setup

We observe a panel of N time series $\{Y_{i,t}\}$ for $i = 1, \dots, N$ and $t = 1, \dots, T$. The first M periods form the calibration sample, assumed to be free of breaks. From period $M + 1$ onward the monitor checks, one period at a time, whether the panel has shifted.

The null and alternative hypotheses are

$$H_0 : \text{no break for any } t > M,$$

$$H_1 : \text{all } N \text{ series shift by } \delta \neq 0 \text{ at some time } \tau > M.$$

Sections 2–5 describe the four steps of the procedure: data generation, AR filtering, cross-sectional aggregation, and the CUSQ test. Section 7 describes the Monte Carlo design, and Section 8 documents the computational improvements.

2 Data Generating Processes

2.1 Panel series

Each series follows an ARFIMA(p_i, d_i, q_i) model,

$$\Phi_i(L)(1-L)^{d_i}Y_{i,t} = \Theta_i(L)\varepsilon_{i,t}, \quad \varepsilon_{i,t} \sim \text{i.i.d. } N(0, 1),$$

where $\Phi_i(L) = 1 - \phi_i L$ and $\Theta_i(L) = 1 + \theta_i L$ are at most first-order polynomials and $(1-L)^{d_i}$ is the fractional differencing operator. Two panel sizes are used.

$N = 10$. Five pairs of series, each pair sharing the same parameters (Table 1).

Table 1: DGP for $N = 10$.				
Series	Model	ϕ_i	d_i	θ_i
1, 2	ARMA(1, 1)	0.8	0	0.5
3, 4	ARMA(1, 1)	0.5	0	0.3
5, 6	ARFIMA(0, d , 0)	0	0.4	0
7, 8	ARFIMA(1, d , 1)	0.1	0.3	0.8
9, 10	ARFIMA(1, d , 0)	0.7	0.4	0

$N = 30$. A random-coefficient panel divided into four blocks (Table 2). The coefficients are drawn independently for each series and for each Monte Carlo replication. This means the simulation averages over both the randomness in the coefficients and the randomness in the innovations, giving a measure of average performance across the whole class of DGPs described in the table rather than performance at one fixed parameter vector.

Table 2: DGP for $N = 30$ (random coefficients, redrawn each replication).

Series	Model	ϕ_i	d_i	θ_i
1–15	ARMA(1, 1)	$U(0, 0.5)$	0	0.6
16–20	ARMA(1, 1)	$U(0.7, 0.9)$	0	0.6
21–25	ARFIMA(0, d , 0)	0	$U(0, 0.5)$	0
26–30	ARFIMA(1, d , 1)	$U(0, 0.4)$	$U(0, 0.5)$	$U(0, 0.3)$

2.2 Common factors

Four common factors are generated independently of the panel series (Table 3). They cover i.i.d. noise, a near-unit-root AR(1), and two long-memory processes.

Table 3: Common factor DGPs.

Factor	Model	ϕ	d
1	i.i.d. $N(0, 1)$	0	0
2	AR(1)	0.99	0
3	ARFIMA(0, d , 0)	0	0.20
4	ARFIMA(0, d , 0)	0	0.45

2.3 Observed data and break injection

The observed series add a common factor with loading $\lambda = 0.5$:

$$Y_{i,t}^* = Y_{i,t} + 0.5 f_t, \quad i = 1, \dots, N. \quad (1)$$

Under H_1 , a level shift of size δ is added at time τ :

$$Y_{i,t}^{**} = Y_{i,t}^* + \delta \cdot \mathbf{1}\{t \geq \tau\}, \quad i = 1, \dots, N. \quad (2)$$

The break time τ is specified in Section 7.2.

3 AR Pre-Filtering

For each series i , we fit an AR(p) model on the calibration sample $t = 1, \dots, M$, where

$$p = \max\left(3, 3 \left\lfloor \frac{M-1}{100} \right\rfloor\right). \quad (3)$$

The OLS estimate from the calibration sample is

$$\hat{\beta}_{AR} = \arg \min_{b \in \mathbb{R}^p} \sum_{t=p+1}^M \left(Y_{i,t} - b'(Y_{i,t-1}, \dots, Y_{i,t-p})' \right)^2. \quad (4)$$

This estimate is fixed after period M and never updated. For all $t = p+1, \dots, T$, the one-step-ahead residual is

$$\hat{e}_{i,t} = Y_{i,t} - \hat{\beta}_{AR}'(Y_{i,t-1}, \dots, Y_{i,t-p})'. \quad (5)$$

The first seven observations are discarded as a burn-in. Fixing the AR coefficients after calibration is the key sequential monitoring requirement: parameters are estimated from clean data and then held constant, so the monitoring period residuals are not used to update the model.

4 Cross-Sectional Aggregation

4.1 Pooled pairwise regression

Let $\hat{e}_{i,t}$ denote the AR residuals for $i = 1, \dots, N$ and $t = 1, \dots, T$. For every ordered pair (i, j) with $i \neq j$ (there are $N(N-1)$ such pairs), we consider the calibration-period regression

$$\hat{e}_{i,t} = \alpha + \beta \hat{e}_{j,t} + u_{i,j,t}, \quad t = 1, \dots, M. \quad (6)$$

A single coefficient pair $(\hat{\alpha}, \hat{\beta})$ is estimated by OLS on the stacked sample of all $N(N-1) \times M$ observations. For all t , the residual from this regression is

$$U_{i,j,t} = \hat{e}_{i,t} - \hat{\alpha} - \hat{\beta} \hat{e}_{j,t}. \quad (7)$$

The aggregate signal used in the CUSQ test is the average of $U_{i,j,t}$ over all ordered pairs:

$$\bar{e}_t \equiv \frac{1}{N(N-1)} \sum_{\substack{i,j=1 \\ i \neq j}}^N U_{i,j,t}, \quad t = 1, \dots, T. \quad (8)$$

As stated, computing (8) requires forming all $N(N-1)$ residual sequences. The next two propositions show that $\hat{\alpha}$, $\hat{\beta}$, and $\bar{e}_t$ all have closed forms that cost $O(NM)$ and $O(NT)$ to evaluate, without any approximation.

4.2 Closed form for $(\hat{\alpha}, \hat{\beta})$

Define the grand calibration mean

$$\mu \equiv \frac{1}{NM} \sum_{i=1}^N \sum_{t=1}^M \hat{e}_{i,t}, \quad (9)$$

demeaned residuals $d_{i,t} \equiv \hat{e}_{i,t} - \mu$ for $t \leq M$, and the cross-sectional sum and sum of squares at each t :

$$D_t \equiv \sum_{i=1}^N d_{i,t}, \quad Q_t \equiv \sum_{i=1}^N d_{i,t}^2, \quad t = 1, \dots, M. \quad (10)$$

Proposition 1. *The pooled OLS estimates from (6) are*

$$\hat{\beta} = \frac{\sum_{t=1}^M (D_t^2 - Q_t)}{(N-1) \sum_{t=1}^M Q_t}, \quad \hat{\alpha} = \mu (1 - \hat{\beta}). \quad (11)$$

Proof. The pooled OLS slope is the ratio of the pooled covariance to the pooled variance. Since each series i appears as the dependent variable in exactly $N - 1$ pairs, both pooled means equal μ . The pooled covariance numerator is $\sum_{t=1}^M \sum_{i \neq j} d_{i,t} d_{j,t}$. Using the identity $\sum_{i \neq j} d_{i,t} d_{j,t} = D_t^2 - Q_t$, this equals $\sum_{t=1}^M (D_t^2 - Q_t)$. Each regressor $\hat{e}_{j,t}$ appears in $N - 1$ pairs as the regressor, so the pooled variance denominator is $(N - 1) \sum_{t=1}^M Q_t$. Taking the ratio gives $\hat{\beta}$. The intercept follows from $\hat{\alpha} = \mu - \hat{\beta} \mu = \mu(1 - \hat{\beta})$. $\square$

4.3 Closed form for $\bar{e}_t$

Proposition 2. *For every $t = 1, \dots, T$,*

$$\bar{e}_t = (1 - \hat{\beta})(\bar{\hat{e}}_t - \mu), \quad \bar{\hat{e}}_t \equiv \frac{1}{N} \sum_{i=1}^N \hat{e}_{i,t}. \quad (12)$$

Proof. Let $S_t = N\bar{\hat{e}}_t$. Expanding (8) using (7):

$$\bar{e}_t = \frac{1}{N(N-1)} \sum_{j=1}^N \sum_{i \neq j} (\hat{e}_{i,t} - \hat{\alpha} - \hat{\beta} \hat{e}_{j,t}).$$

For the first term, $\sum_j \sum_{i \neq j} \hat{e}_{i,t} = \sum_j (S_t - \hat{e}_{j,t}) = (N-1)S_t$. For the second, $\sum_j \sum_{i \neq j} \hat{\alpha} = N(N-1)\hat{\alpha}$. For the third, $\sum_j \sum_{i \neq j} \hat{\beta} \hat{e}_{j,t} = (N-1)\hat{\beta}S_t$. Therefore

$$\bar{e}_t = \frac{(N-1)S_t - N(N-1)\hat{\alpha} - (N-1)\hat{\beta}S_t}{N(N-1)} = (1 - \hat{\beta})\bar{\hat{e}}_t - \hat{\alpha}.$$

Substituting $\hat{\alpha} = \mu(1 - \hat{\beta})$ gives (12). $\square$

Remark 1. *Proposition 2 says that, because all $N(N-1)$ pairwise regressions share the same $(\hat{\alpha}, \hat{\beta})$, averaging their residuals is the same as (i) computing the plain cross-sectional mean $\bar{\hat{e}}_t$ and (ii) applying the affine map $(1 - \hat{\beta})(\cdot - \mu)$. The total cost is $O(NM)$ to estimate the coefficients and $O(NT)$ to compute $\bar{e}_t$ for the full sample. For $N = 30$ this replaces an $N(N-1) = 870$ -pair construction with a single pass over $N = 30$ series.*

5 The CUSQ Statistic and Monitoring Boundary

5.1 Statistic

Let $C_n = \sum_{t=1}^n \bar{e}_t^2$ be the cumulative sum of squares of $\bar{e}_t$. Following Wang and Hsiao (2013), define for $n > M$:

$$D_n \equiv \frac{C_n}{C_M} - \frac{n}{M}, \quad (13)$$

$$T_n \equiv \sqrt{\frac{M}{2}} D_n = \sqrt{\frac{M}{2}} \left(\frac{C_n}{C_M} - \frac{n}{M} \right). \quad (14)$$

The centering term is n/M , not the $(n-2)/(M-2)$ correction of Inclán and Tiao (1994). The two are asymptotically the same, but n/M is the form used by Wang and Hsiao (2013) and is the one for which the Brownian bridge limit below holds as stated.

5.2 Limiting distribution

Under H_0 , Wang and Hsiao (2013) (Theorem 1) show that

$$T_n \Rightarrow W^0(\tau), \quad \tau = \frac{n}{M}, \quad (15)$$

where $W^0(\tau) = W(\tau) - \tau W(1)$ is a Brownian bridge. The probability that $|W^0(\tau)|/\sqrt{\tau}$ exceeds a threshold a for some $\tau \geq 1$ is (Chu et al., 1996):

$$\Pr\left(\sup_{\tau \geq 1} \frac{|W^0(\tau)|}{\sqrt{\tau}} \geq a\right) = 2[1 - \Phi(a)] + 2a\phi(a) = \alpha, \quad (16)$$

where Φ and ϕ are the standard-normal CDF and PDF. Solving (16) numerically gives

$$a_{10\%}^2 = 6.2514, \quad a_{5\%}^2 = 7.8147, \quad a_{1\%}^2 = 11.3449. \quad (17)$$

Wang and Hsiao (2013) quote $a_{5\%}^2 = 7.78$, which is the value rounded to two decimal places; the exact solution is 7.8147.

5.3 Monitoring boundary

The boundary that controls the crossing probability at level α is (Wang and Hsiao, 2013):

$$g(\tau; a_\alpha^2) = \sqrt{\tau(a_\alpha^2 + \ln \tau)}, \quad \tau = \frac{n}{M}. \quad (18)$$

This differs from the boundary of Chu et al. (1996),

$$g_{\text{CSW}}(\tau) = (\tau - 1) \sqrt{\frac{\tau}{\tau - 1} \left(a^2 + \ln \frac{\tau}{\tau - 1} \right)},$$

which arises from a different parametrisation of the same Wiener process. The two boundaries are equal only at $\tau = 2$. At $\tau = 1.1$, g_{CSW} is about three times smaller than (18); at $\tau = 5$ it is about twice as large. Using the wrong boundary therefore has a substantial effect on size and power at most monitoring horizons.

6 Sequential Decision Rule

At each period $n > M$, compute T_n from (14) and compare it to the boundary $g(\tau; a_\alpha^2)$ from (18) with $\tau = n/M$. The procedure declares a break at the first period where the statistic crosses the boundary:

$$\hat{\tau}_\alpha \equiv \inf\{n > M : |T_n| \geq g(n/M; a_\alpha^2)\}, \quad (19)$$

with $\hat{\tau}_\alpha = +\infty$ if no crossing occurs before T . The estimates $(\hat{\beta}_{AR}, \hat{\alpha}, \hat{\beta}, \mu)$ are all fixed from the calibration period and never updated after $t = M$.

7 Monte Carlo Design

The total sample length used in each replication is

$$T_{\text{total}} = 10M + 10, \quad (20)$$

which provides a monitoring window of $9M$ periods beyond the calibration sample. Two scenarios are run.

7.1 Scenario A: size ($\delta = 0$)

No break is injected. Because the data do not depend on k , each simulated path yields one stopping time $\hat{\tau}_\alpha^{(r)}$, and this is scored against six window lengths $k \in \mathcal{K} = \{0.2, 0.3, 0.5, 1, 2, 3\}$ at once:

$$A_\alpha^{(r)}(k) \equiv \mathbf{1}\{\hat{\tau}_\alpha^{(r)} \leq M + kM\}. \quad (21)$$

The empirical size is

$$\widehat{\text{Size}}_\alpha(N, M, \text{factor}, k) = \frac{1}{R} \sum_{r=1}^R A_\alpha^{(r)}(k), \quad (22)$$

which estimates the false alarm probability within the first kM monitoring periods. A well-calibrated test should give $\widehat{\text{Size}}_\alpha \approx \alpha$.

7.2 Scenario B: power ($\delta \in \{1, 2, 3\}$)

A break of size δ is injected at

$$\tau(k) = M + \lfloor kM \rfloor, \quad k \in \mathcal{K}. \quad (23)$$

Here k is the break location, not a scoring window. Each (δ, k) combination requires its own set of replications because the data depend on (δ, k) . The stopping time $\hat{\tau}_\alpha^{(r)}$ is compared to

the true break date $\tau(k)$ and classified as one of three outcomes:

$$P_\alpha^{(r)} \equiv \mathbf{1}\{\hat{\tau}_\alpha^{(r)} \leq \tau(k)\} \quad (\text{alarm before the break}), \quad (24)$$

$$D_\alpha^{(r)} \equiv \mathbf{1}\{\tau(k) < \hat{\tau}_\alpha^{(r)} < \infty\} \quad (\text{alarm after the break}), \quad (25)$$

$$B_\alpha^{(r)} \equiv P_\alpha^{(r)} + D_\alpha^{(r)} = \mathbf{1}\{\hat{\tau}_\alpha^{(r)} < \infty\} \quad (\text{any alarm}). \quad (26)$$

The Monte Carlo averages are

$$\hat{P}_\alpha = \frac{1}{R} \sum_r P_\alpha^{(r)}, \quad \hat{D}_\alpha = \frac{1}{R} \sum_r D_\alpha^{(r)}, \quad \hat{B}_\alpha = \hat{P}_\alpha + \hat{D}_\alpha. \quad (27)$$

$\hat{D}_\alpha$ is the power of the test — the probability of detecting the break after it occurs. $\hat{P}_\alpha$ is the premature alarm rate.

For replications where $D_\alpha^{(r)} = 1$, the detection delay is

$$\text{Delay}_\alpha^{(r)} \equiv \hat{\tau}_\alpha^{(r)} - \tau(k) > 0. \quad (28)$$

Summary statistics (mean, standard deviation, median, 25th and 75th percentiles) are computed over the subset $\{r : D_\alpha^{(r)} = 1\}$. Premature alarms ($P_\alpha^{(r)} = 1$) are excluded from this calculation because they have a negative delay and measure a different phenomenon.

7.3 Full design

Table 4 lists all parameter combinations.

Table 4: Monte Carlo design.

Parameter	Scenario A (size)	Scenario B (power)
N	$\{10, 30\}$	$\{10, 30\}$
δ	0	$\{1, 2, 3\}$
k	scoring window, $\{0.2, 0.3, 0.5, 1, 2, 3\}$	break location, $\{0.2, 0.3, 0.5, 1, 2, 3\}$
M	$\{50, 80, 100, 200, 300\}$	$\{50, 80, 100, 200, 300\}$
Factor	$\{1, 2, 3, 4\}$	$\{1, 2, 3, 4\}$
R	2500	2500
Combinations	$2 \times 5 \times 4 = 40$	$2 \times 3 \times 6 \times 5 \times 4 = 720$
Total replications	10^5	1.8×10^6

For $N = 30$, the random coefficients in Table 2 are redrawn on every replication, so the Monte Carlo averages integrate over both the coefficient distribution and the innovation sequences.

8 Computational Details

The full design requires 1.9×10^6 replications. Two changes to the implementation reduce computation time substantially without changing any statistic.

8.1 FFT-based fractional differencing

An ARFIMA process requires applying the fractional integration operator $(1 - L)^{-d}$, which has the moving-average representation

$$(1 - L)^{-d} = \sum_{j=0}^{\infty} w_j L^j, \quad w_j = \frac{\Gamma(j + d)}{\Gamma(j + 1) \Gamma(d)}. \quad (29)$$

Given a short-memory series Y_t , the fractionally integrated series is

$$X_t = \sum_{j=0}^t w_j Y_{t-j}, \quad t = 0, 1, \dots, T - 1. \quad (30)$$

Computing (30) by a direct loop costs $O(T^2)$ operations. The same result can be obtained by the discrete convolution

$$(w * Y) = \mathcal{F}^{-1}[\mathcal{F}(w) \odot \mathcal{F}(Y)], \quad (31)$$

where $\mathcal{F}$ is the DFT computed by the FFT algorithm, at cost $O(T \log T)$. The first T elements of $(w * Y)$ equal (30) exactly. For numerical stability, the weights are computed in log space:

$$w_j = \exp[\ln \Gamma(j + d) - \ln \Gamma(j + 1) - \ln \Gamma(d)], \quad (32)$$

which avoids overflow for large j . For $T = 4000$, this reduces the fractional differencing step from hundreds of milliseconds to under one millisecond per series.

8.2 Vectorized AR residuals

Equation (5) computes one residual per time period. A direct implementation loops over t and computes a dot product at each step, incurring $T - p$ separate Python function call overheads. The same result follows from a single matrix multiplication.

Lemma 1. *Let $t_0 = \max(8, p + 1)$ and $\mathcal{I} = \{t_0, \dots, T\}$. Define the lag matrix $Z \in \mathbb{R}^{|\mathcal{I}| \times p}$ by*

$$Z_{t,\ell} = Y_{i,t-\ell}, \quad \ell = 1, \dots, p, \quad t \in \mathcal{I}, \quad (33)$$

and let $y = (Y_{i,t})_{t \in \mathcal{I}}$. Then

$$(\hat{e}_{i,t})_{t \in \mathcal{I}} = y - Z \hat{\beta}_{AR}. \quad (34)$$

Proof. Row t of $Z \hat{\beta}_{AR}$ equals $\sum_{\ell=1}^p \hat{\beta}_{AR,\ell} Y_{i,t-\ell}$, which is the predicted value in (5). So $(y - Z \hat{\beta}_{AR})_t = \hat{e}_{i,t}$ for every $t \in \mathcal{I}$. $\square$

The matrix Z is formed once and the residuals for all periods are obtained in a single BLAS call. For $T \approx 3000$ and $p \leq 6$, this is about 30 times faster than the loop version, with identical numerical output.

8.3 Parallelisation

After both improvements, the time for one replication at $N = 30$, $M = 300$ falls from roughly 0.14 seconds to roughly 0.02 seconds, a reduction of about $8\times$. Across 1.9×10^6 replications, this brings the projected single-thread runtime from about 72 hours to about 9.5 hours.

The 760 parameter combinations in Table 4 are independent of one another, so each combination is assigned to one of 8 worker processes. Each worker is seeded from OS entropy to ensure independent random streams across processes. With 8-way parallelism, the expected wall-clock time is 1–1.5 hours.

9 Summary

Scenario A produces 240 rows (40 combinations $\times$ 6 values of k), each reporting $\widehat{\text{Size}}_\alpha$ for $\alpha \in \{10\%, 5\%, 1\%\}$. Scenario B produces 720 rows, each reporting $\widehat{B}_\alpha$, $\widehat{P}_\alpha$, $\widehat{D}_\alpha$, and the five delay statistics for each α .

In every replication, the CUSQ statistic uses $D_n = C_n/C_M - n/M$ from (13) and the boundary $g(\tau) = \sqrt{\tau(a_\alpha^2 + \ln \tau)}$ from (18), both as specified in Wang and Hsiao (2013). Propositions 1 and 2 ensure that the pooled cross-pair aggregate $\bar{e}_t$ is computed exactly, at cost $O(NT)$ instead of $O(N^2T)$. Lemma 1 and the FFT-based fractional-differencing step ensure that all residuals are numerically identical to a direct implementation.

References

- Brown, R. L., J. Durbin, and J. M. Evans (1975). Techniques for testing the constancy of regression relationships over time. *Journal of the Royal Statistical Society, Series B*, 37, 149–163.
- Chu, C.-S. J., M. Stinchcombe, and H. White (1996). Monitoring structural change. *Econometrica*, 64(5), 1045–1065.
- Inclan, C. and G. C. Tiao (1994). Use of cumulative sums of squares for retrospective detection of changes of variance. *Journal of the American Statistical Association*, 89(427), 913–923.
- Poskitt, D. S. (2007). Autoregressive approximation in nonstandard situations: the fractionally integrated and non-invertible cases. *Annals of the Institute of Statistical Mathematics*, 59, 697–725.
- Wang, C. S.-H. and C. Hsiao (2013). Real-time monitoring test for realized volatility. *Journal of Time Series Econometrics*, 5(1), 1–24.